

WASP-12b

R-1631

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_191106 · created 2026-08-02T05:46:49+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458793.88904 +/- 0.00068 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.068 +/- 0.046
Transit depth (Rp/Rs)^2	0.46 +/- 0.62 [%]
Orbital Inclination (inc)	77.25 +/- 2.86
Airmass coefficient 1 (a1)	0.18 +/- 0.11
Airmass coefficient 2 (a2)	1.57 +/- 0.89
Scatter in the residuals of the lightcurve fit is	59.05 %
Best Comparison Star	None
Optimal Aperture	3.09
Optimal Annulus	9.61
Transit Duration (day)	0.096 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.068 ± 0.046 vs 0.1178 (deviation 1.08σ)
Duration fit vs published	0.096 d vs 0.125 d (deviation 1.71σ)
Mid-time O–C	-1.9 min
Depth SNR	1.5
Residual scatter	59.05 %

Light curve

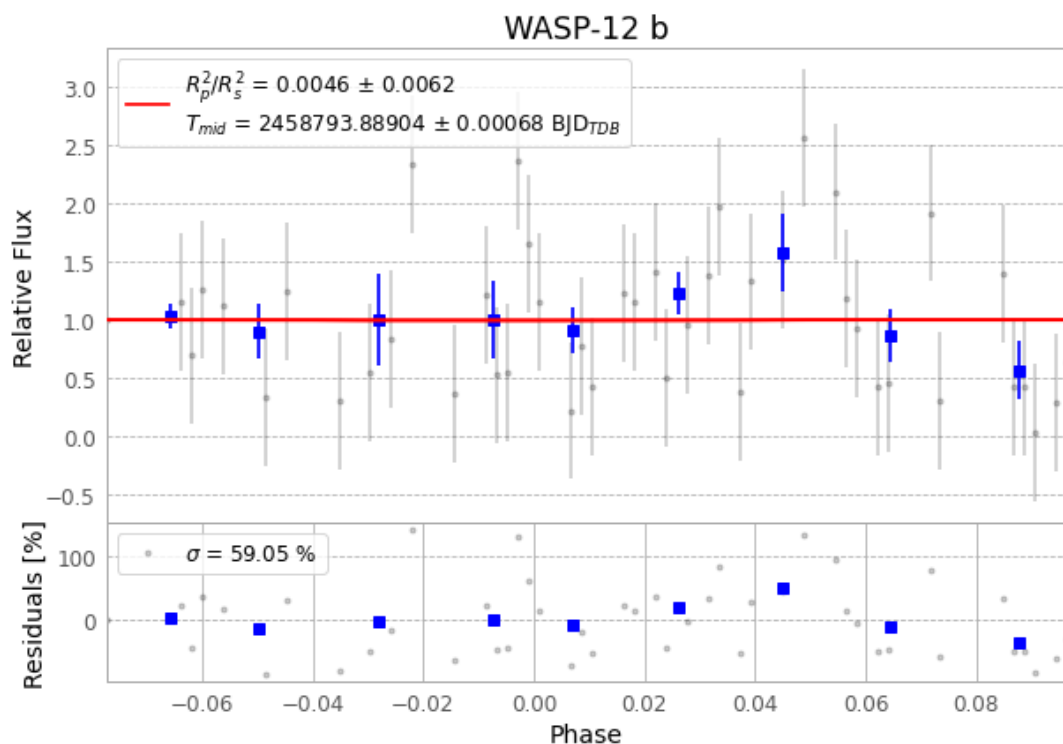


Figure 1 — FinalLightCurve_WASP-12 b_2019-11-06.png (SHA-256 4cc70e694a3029c1....)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [317, 105], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3ce745b8cf163848...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

92 FITS frames (60 MB), WASP-12191106071212.FITS → WASP-12191106114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-191106.log (4dca6839f0cf...), FinalLightCurve_WASP-12 b_2019-11-06.png (4cc70e694a30...), FinalLightCurve_WASP-12 b_2019-11-06.csv (d57f981ba34d...)

Compute resources

Wall time 200.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 05:46Z.